Patterns at the workbench

Tungsten Carbide

Name _____ Date _____

Show the equal steps with numbers or small jumps.

1. Little counts pens with tiny balls in pairs. What count comes next?

2, 4, 6, ___

Show your thinking.

2. A closed drill-bit case has slots grouped in fives. What slot count comes next?

5, 10, 15, ___

Show your thinking.

3. Little counts drawn saw teeth in tens. What count comes next?

10, 20, 30, ___

Show your thinking.

Patterns at the workbench

Tungsten Carbide

Name _____ Date _____

Show the equal steps with numbers or small jumps.

4. Little counts pole marks along a snowy path in twos. What count is missing?

12, ___, 16, 18

Show your thinking.

5. Little groups scratch-test notes in fives. What count is missing?

20, 25, ___, 35

Show your thinking.

6. A workshop notebook counts packed pens in tens. What count is missing?

60, ___, 80, 90

Show your thinking.

WONDER DAILY

Tungsten Carbide

A little book of very hard things



Very hard. Not always tough.

GRADE 3

A gray tip rested on a cloth. "Could that little piece do a big job?" asked Little Wonderer. "It is tungsten carbide," said Big. "A material is what something is made of. This one is heavy for its size and very hard. But being hard does not mean it can never break."



Little found another clue in a pen. Many ballpoint pens have a tiny tungsten-carbide ball that rolls ink onto paper. People make carbide by joining tungsten, a metal, with carbon. They press the powder and heat it very hot. A little cobalt metal usually holds the grains together, like glue.



Big set soap and a wooden craft stick on a tray. Little guessed what would happen, then watched. Big gently drew the stick across the soap. A scratch appeared: a small line dug into the surface. Hard means resisting scratches and dents. The soap gave way to the harder stick.

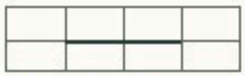


Does every material bend before it breaks? Big gently bent chalk over the tray. It snapped: chalk is brittle. A rubber eraser bent and came back: it is tough. Tough means resisting cracks and breaks from bumps or bends. Tungsten carbide is very hard, but brittle, so a strong knock can chip it.



Make a little book.

1. Print one-sided: US Letter, portrait, actual size / 100%. No fit. Headers/footers off.
2. Fold long edge to long edge; reopen. Fold short edge to short edge, then in half again. Open: eight boxes.
3. Cut only the solid center line across the middle two boxes. Keep the outer boxes joined.
4. Refold long edge to long edge, print outside. Push ends together to open a diamond, then a cross.
5. Wrap into a book with page 1 in front. Follow pages 1 to 8.



Predict, watch a grown-up test, and pause whenever you like.

Your turn

Write one command for the soap test.

Example: Ask Big to hold the soap and gently drag the craft stick across it.



Back at the notebook, Big showed a saw-tooth picture. A small carbide tip joined a steel body. The hard tip resists wear; the tougher steel takes bumps without cracking as easily. "Two materials, two jobs," Little said. "Do I need it hard, tough, or both?"



On a snowy walk, Little noticed the tips of their hiking poles. The tip is the small end that meets the ground. Some pole tips and ice-grip studs use carbide to bite into ice. "The hard part is only a tiny piece!" Little said. "Just where it is useful," said Big.

Give the test its words

Tungsten Carbide · Curiosity day

Name _____ Date _____

Write one command about the soap test. Say it first if you like.



A line in the soap



A bend and a break



Hard, tough, or both

Draw the scratch Big made in the soap. Add a label.

ask · watch · record · soap · scratch · craft stick · hard · tough

Ask Big to … name one gentle step with the soap and craft stick

Watch the … tell what to notice while Big leads the test

Record the … tell what mark to save in the notebook

Does your sentence tell what to do? Keep one clear command.